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Factors Affecting Performances of Small Projects in Small Island States – Stakeholders’ Perceptions

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Abstract. Numerous are the studies that investigate the performances of big construction projects. However, in this particular investigation, the emphasis is on small construction projects. The intent was to capture the perceptions of the major stakeholders, that is the clients, contractors and consultants on the factors that could affect the performances of such small projects, which have limited budget, are of short duration and to further complex the situation, the projects are executed in small island states, such as the Republic of Mauritius. After an extensive literature search, a list of 61 performance indicators was made, which was further grouped into 10 main clusters, namely (i) cost; (ii) time; (iii) quality; (iv) productivity; (v) owner; (vi) regulator; (vii) employee; (viii) health and safety; (ix) learning; and (x) environment. A performance success survey was devised, tested and refined before circulating to the major stakeholders (response rate; client – 89%; contractors – 68.5% and consultants – 56%). With Kendall’s coefficient of concordance $W = 0.954$ being greater than 0.5, the degree of agreement amongst stakeholders was found to be statistically significant. Hence the major 5 factors that impeded on the performances of smaller construction projects being executed in small island countries were found to be due to (i) poor leadership skills; (ii) absence of skilled force; (iii) inadequate liquidity within the company; (iv) escalation of material prices and (v) unavailability of resources.

1. Introduction

The economic growth of any nation rests on many pillars, and one of them is the construction industry, which through the development of physical infrastructures, such as buildings, roads and bridges among others, indicates the country’s financial evolution [1]. In the construction sector, consultants, such as project managers, architects, engineers, cost estimators among others, are appointed to ensure that the project is designed to prevailing codes of practice and executed accordingly. During the construction phase, the professionals make sure that the selected contractor carries out the works according to plans and specifications. They all work towards similar goals, that of completing the project on time, within the allocated budget and of quality matching industry standard, which are the commonest indicators for construction success [2,3,4, 5, 6, 7].

The literature has extensive works carried out by researchers that show how the performances on major construction projects are getting worse with time. In the United Arab Emirates, failures of many projects have been attributed to inexperienced workforce, inadequate supervision of the works, poor site management, incompetent leaders among the teams, lack and poorly serviced equipment that most of the time are not in use due to breakdowns [8]. According to Ajayi et al. [9], the contractor(s) selected has a substantial effect on the success or failure of the projects. A study by Puspasari [3] acknowledged 46 factors, which he categorised under eight groups, that could be held responsible for the poor performance of such construction ventures, namely; clients’ issues, contractors’ issues, consultants’ issues, subcontractors’ issues, factors related to material and labour, contractual relationship factors, project procedures and external environment factors. The poor performances on such major projects do happen despite the appointment and supervision by a team of consultants.



Compared to major construction projects, small ones [10] have a shorter lifespan, higher ambiguity and limited formal documentation. Repetitive jobs and simple construction process are key attributes of small projects [11]. Projects were categorised as being small whenever the project value ranged between US\$0.1 million and US \$5 million [12].

Most of the research done concerning gauging the performances of construction projects have been oriented towards big countries (developed and industrialised ones). The performances of small projects in small island states have never been looked at. Briguglio in his publication of 1995 [13] describes how Small Island Developing States (SIDS), due to their trivial extent, insularity, inaccessibility and proneness to natural catastrophes, face a lot of problems that put their economies at stake, which is reflected in either their Gross Domestic Product (GDP) or Gross National Product (GNP) per capital. Smallness is undeniably an attribute of SIDS, measured in terms of its number of inhabitants, area of land and/or its GNP, which makes that particular state economically disadvantageous for the reasons listed hereunder [14,15, 16, 17, 18]

- The natural resources are limited making the economy dependent on imports.
- The resulting protected economic environment following the adoption of the import-substitution policies had products of mediocre class, upper charges and also created a comparable market in non-domestically produced goods.
- The local marketplace is small, and high affinity on export arcades exist.
- Inability to diversify its exports makes the nation dependent on a limited number of goods and services.
- Inadequate capacity to impact on national fees.
- Inadequate capability to exploit economies of scale.
- Insufficient local opposition as the state sustains numerous companies with similar product.
- Workforce - experienced and efficient administrators are limited.

Furthermore, all islands are insular, and by their location, the main constraints for these specific nations are the higher per unit transportation cost as all their imports and exports have to use either air or sea transportation. Unreliability in transport services and time delays handicaps the provision of industrial supplies/materials. Natural calamities impact significantly on the island economy due to its small size.

In this particular study, the intent is to investigate the stakeholders' perceptions of the factors that affect performances of small construction projects in small island states; the case study was done for the Republic of Mauritius. The Island of Mauritius, consist of some outer islands with the two main ones being Mauritius and Rodrigues, is located South-West of the Indian Ocean. The main island, Mauritius with a population of 1,219,265 as at 1st July 2016 and an area of 1865 km² [19] is one of the listed Small Island Developing States [20].

2. Methodology

The different stakeholders within the construction sector are illustrated in figure 1. However, it should be made clear that even when occupying the same position, the attributes such as the roles, rights, responsibilities, and principles of action may sometimes differ from nation to nation [21]. In all, there are three major stakeholders, clients, contractors and consultants.

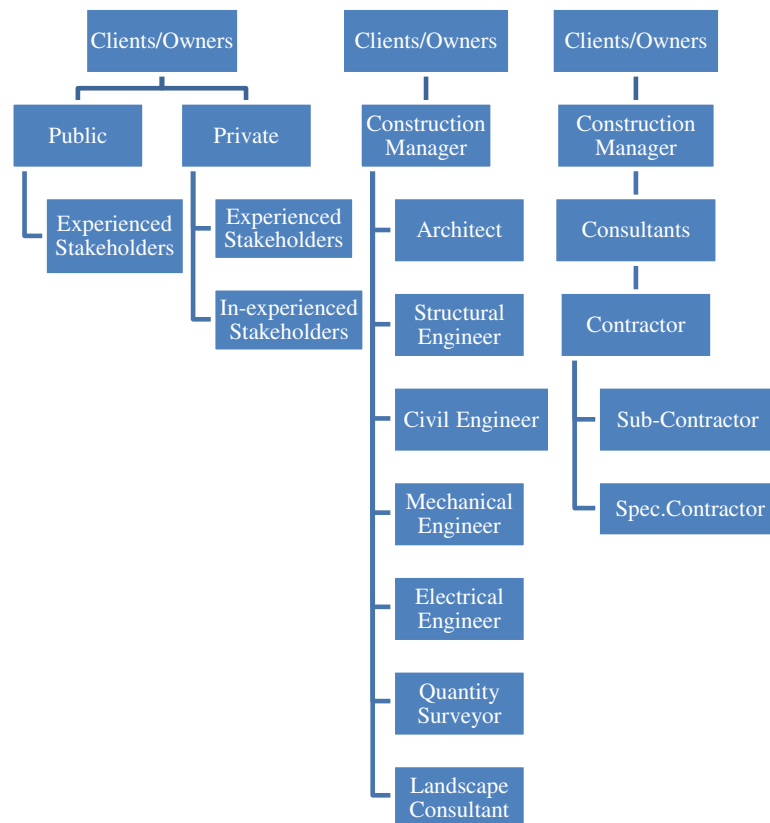


Figure 1. Key Players of the construction industry.

The following steps were followed in the determination of factors affecting construction performance.

Step 1: Conduct a thorough literature review.

Using the different available resources, (internet, library databases, google scholar, science direct and reputable online open access journals among others), an in-depth review of the literature was performed to identify the key factors affecting the performance/success of construction projects in general.

A list of 61 performance indicators, obtained from the works of the following researchers, namely; Ugwu and Haupt [22], Jha and Iyer [23], Love *et al.* [24], Navon [25], Samson and Lema [26], Kuprenas [27], Cheung *et al.* [28], Lehtonen [29], Brown and Adams [30], DETR [31], Dissannayaka and Kumaraswamy [32], Karim and Marosszeky [33], Reichelt and Lyneis [34], were divided into 10 clusters, namely:

- (a) cost,
- (b) time,
- (c) quality,
- (d) productivity,
- (e) owner,
- (f) regulator,
- (g) employee,
- (h) health and safety,
- (i) learning, and
- (j) environment.

Step 2: Development of survey instruments.

Using the outcomes of the intensive literature review, the project performance /success survey questionnaire was devised, part 1 gathered information on both the respondent and his/her firm while part 2 included questions about the ten groups assigned to the performance indicators.

Step 3: Pilot study.

The survey questionnaire was circulated to a team of experts, which comprised;

- (i) a professional architect,
- (ii) two professional engineers,
- (iii) two builders,
- (iv) two clients with experience in construction and
- (v) two quantity surveyors.

After taking on board the different comments and recommendations, the questionnaire was updated.

Step 4: Determination of performance factors from major stakeholders.

The refined survey questionnaire was circulated by emails and post to projects' clients, consultants and contractors to capture their perceptions concerning factors that hinder performance on construction projects. 65 Clients, 50 Consultants and 92 Contractors (all grades included) were contacted. The five-point Likert scale was utilised [1: Not important; 2: Less important; 3: moderate; 4: Important; and 5: Very important] with a mean cut-off score of 3.0, hence all indicators with a mean score less than 3.0 was discarded from the list.

In this particular research work, the relative importance index method (RII) was utilised to determine owners', consultants', and contractors' sensitivities of the relative significance of the acknowledged performance dynamics. The RII was calculated using the formula proposed by Ugwu and Haupt [22], Jha and Iyer [23], and Chueng et al. [28], which is as follows.

$$RII = \frac{\sum W}{AN} \quad (1)$$

where W is the weight given to each factor by the respondents and ranges from 1 to 5; A being the highest weight equal to 5 and N being the total number of participants.

Kendall's coefficient was used to determine the degree of agreement among the three essential groups of respondents on a zero to one scale. Kendall's W is a value between 0 and 1, where 0 indicates no agreement and 1 indicates complete agreement. W is calculated using the equation by Kendall et al. [35].

$$W = \frac{12 \sum D^2}{m^2 N(N^2 - 1)} \quad (2)$$

Where: D = the difference between the individual sum of ranks of the stakeholders and the average of the sum of ranks of the factors

$\sum D^2$ = the sum of the squares of the difference

m = number of the stakeholder's group

N = number of factors being ranked.

The problem is to find if there is an agreement or concordance among the three different stakeholders regarding the 61 performance factors.

Null hypothesis (H0: There is an insignificant degree of agreement among Clients, Contractors and Consultants) and the alternative hypothesis (H1: There is a statistically significant degree of consensus among owners, contractors and consultants) were tested.

3. Results and discussions

Despite the busy schedule of the major stakeholders, 89% of the clients; 68.5% of the contractors and 56% of the consultants responded to the survey.

Tables 1 to 3 illustrate the results obtained from the survey of major stakeholders regarding the factors affecting the performance of construction projects. Each chart provides information about each factor's RII, mean score and overall rank.

Table 1. RII and ORDER of 61 factors – Clients' perceptions.

		PERFORMANCE FEATURES	Rank	Weighted total	RII	Item Mean
1	COST FEATURES	Company's market share	61	58	0.200	1.000
2		Liquidity within company	2	240	0.828	4.138
3		Project - Cash flow	32	130	0.448	2.241
4		Project - Profit rate	29	135	0.466	2.328
5		Project - Overheads	23	146	0.503	2.517
6		Project - design cost	27	139	0.479	2.397
7		Material and equipment cost	31	132	0.455	2.276
8		Project - labour cost	33	130	0.448	2.241
9		Project - over time cost	21	153	0.528	2.638
10		Stimulus Package – Cost/budget	24	144	0.497	2.483
11		Re-work - Cost	5	225	0.776	3.879
12		Variations	6	221	0.762	3.810
13		Wastage - Materials	11	210	0.724	3.621
14		Regular project budget update	34	130	0.448	2.241
15		Cost control method	47	100	0.345	1.724
16		Escalation - material prices	13	206	0.710	3.552
17		Currency prices - Differentiation	30	134	0.462	2.310
18	TIME FEATURES	Time - Site preparation	22	148	0.510	2.552
19		Planned construction time	17	192	0.662	3.310
20		Late deliveries - Percentage	25	143	0.493	2.466
21		Variation orders – Implementation time	7	220	0.759	3.793
22		Defects rectification – Time required	8	219	0.755	3.776
23		Claim approval - Delays	9	213	0.734	3.672
24		Regular payments – Average delays	10	211	0.728	3.638
25		Unavailability of resources	4	237	0.817	4.086
26		Average delay because of closures leading to materials shortage	60	59	0.203	1.017
27		Conformity with specification	45	101	0.348	1.741
28	QUALITY	The absence of a skilled workforce	3	239	0.824	4.121
29		Quality - equipment and raw materials	14	202	0.697	3.483
30		Quality assessment method in company	36	126	0.434	2.172
31		Quality training/meeting	50	98	0.338	1.690
32	EFFICIENCY	Difficulties met during the project	26	143	0.493	2.466
33		New projects rate per annum	37	123	0.424	2.121
34		Relationship between Management & labour	42	103	0.355	1.776
35		Absence rate on Project	12	207	0.714	3.569
36	OWNER	Work sequence with respect to programme	15	198	0.683	3.414
37		Coordination bet. owner and other stakeholders	28	136	0.469	2.345
38		PM – leadership skills	1	248	0.855	4.276
39		Speed and reliability of service	35	129	0.445	2.224
40	COMPLIANCE	Disagreements – Client & project stakeholders	51	97	0.334	1.672
41		Amount of re-works	16	191	0.682	3.411
42		Compliance cost (Authorities)	59	72	0.248	1.241
43		Frequency - Non-compliant cases	57	79	0.272	1.362
44	EMPLOYEE	Regulatory documents – Quality & Readiness	19	190	0.655	3.276
45		Issues – Site conditions	38	107	0.369	1.845
46		Workers' defiance	39	105	0.362	1.810
47		Employment and skill improvement	40	103	0.355	1.776
48	H&S	Workers stimulus	58	79	0.272	1.362
49		Sense of belonging to work	18	191	0.659	3.293
50		HOS - Application in organization	56	86	0.297	1.483
51		Project place - safe to reach	48	100	0.345	1.724
52	LEARNING	Project accident rate – (Reportable ones only)	49	98	0.338	1.690
53		Project Assurance rate	54	87	0.300	1.500
54		Learning (individual practice and former history)	52	94	0.324	1.621
55		Learning (best practice & exp. of others)	43	101	0.348	1.741
56	ENVIRONMENT	Work group	55	87	0.300	1.500
57		Appraisal of failures & Provision of solution	20	190	0.655	3.276
58		Quality of Air	53	93	0.321	1.603
59		Level of noise	44	101	0.348	1.741
60		Trashes	46	101	0.348	1.741
61		Climate condition	41	103	0.355	1.776

Table 2. RII and ORDER of 61 factors – Consultants’ perceptions.

		PERFORMANCE FEATURES	Rank	Weighted total	RII	Item Mean
1	COST FEATURES	Company’s market share	61	41	0.293	1.464
2		Liquidity within company	3	125	0.893	4.464
3		Project - Cash flow	33	90	0.643	3.214
4		Project - Profit rate	30	92	0.657	3.286
5		Project - Overheads	26	94	0.671	3.357
6		Project - design cost	28	94	0.671	3.357
7		Material and equipment cost	32	91	0.650	3.250
8		Project - labour cost	34	90	0.643	3.214
9		Project - over time cost	23	94	0.671	3.357
10		Stimulus Package – Cost/budget	41	87	0.621	3.107
11		Re-work - Cost	7	118	0.843	4.214
12		Variations	12	115	0.821	4.107
13		Wastage - Materials	17	112	0.800	4.000
14		Regular project budget update	35	89	0.636	3.179
15		Cost control method	48	81	0.579	2.893
16		Escalation - material prices	4	124	0.886	4.429
17	TIME FEATURES	Currency prices - Differentiation	31	91	0.650	3.250
18		Time - Site preparation	24	94	0.671	3.357
19		Planned construction time	19	108	0.771	3.857
20		Late deliveries - Percentage	27	94	0.671	3.357
21		Variation orders – Implementation time	13	115	0.821	4.107
22		Defects rectification – Time required	14	114	0.814	4.071
23		Claim approval - Delays	15	113	0.807	4.036
24		Regular payments – Average delays	16	112	0.800	4.000
25		Unavailability of resources	5	122	0.871	4.357
26		Average delay because of closures leading to materials shortage	60	42	0.300	1.500
27	QUALITY	Conformity with specification	21	95	0.679	3.393
28		The absence of a skilled workforce	2	126	0.900	4.500
29		Quality - equipment and raw materials	18	111	0.793	3.964
30		Quality assessment method in company	38	89	0.636	3.179
31	EFFICIENCY	Quality training/meeting	25	94	0.671	3.357
32		Difficulties met during the project	37	89	0.636	3.179
33		New projects rate per annum	39	87	0.621	3.107
34		Relationship between Management & labour	22	95	0.679	3.393
35	OWNER	Absence rate on Project	6	119	0.850	4.250
36		Work sequence with respect to programme	8	118	0.843	4.214
37		Coordination bet. owner and other stakeholders	29	93	0.664	3.321
38		PM – leadership skills	1	128	0.914	4.571
39	COMPLIANCE	Speed and reliability of service	36	89	0.636	3.179
40		Disagreements – Client & project stakeholders	51	76	0.543	2.714
41		Amount of re-works	11	116	0.829	4.143
42		Compliance cost (Authorities)	59	64	0.457	2.286
43	EMPLOYEE	Frequency - Non-compliant cases	57	67	0.479	2.393
44		Regulatory documents – Quality & Readiness	20	107	0.764	3.821
45		Issues – Site conditions	40	87	0.621	3.107
46		Workers’ defiance	42	87	0.621	3.107
47	H&S	Employment and skill improvement	43	86	0.614	3.071
48		Workers stimulus	58	64	0.457	2.286
49		Sense of belonging to work	10	116	0.829	4.143
50		HOS - Application in organization	56	67	0.479	2.393
51	LEARNING	Project place - safe to reach	49	80	0.571	2.857
52		Project accident rate – (Reportable ones only)	50	79	0.564	2.821
53		Project Assurance rate	54	70	0.500	2.500
54		Learning (individual practice and former history)	52	76	0.543	2.714
55	ENVIRONMENT	Learning (best practice & exp. of others)	45	84	0.600	3.000
56		Work group	55	69	0.493	2.464
57		Appraisal of failures & Provision of solution	9	117	0.836	4.179
58		Quality of Air	53	72	0.514	2.571
59		Level of noise	46	84	0.600	3.000
60		Trashes	47	82	0.586	2.929
61		Climate condition	44	85	0.607	3.036

Table 3. RII and ORDER of 61 factors – Contractors' perceptions.

		PERFORMANCE FEATURES	Rank	Weighted total	RII	Item Mean
1	COST FEATURES	Company's market share	22	203	0.644	3.222
2		Liquidity within company	2	276	0.876	4.381
3		Project - Cash flow	36	165	0.524	2.619
4		Project - Profit rate	33	166	0.527	2.635
5		Project - Overheads	29	171	0.543	2.714
6		Project - design cost	31	169	0.537	2.683
7		Material and equipment cost	35	165	0.524	2.619
8		Project - labour cost	37	161	0.511	2.556
9		Project - over time cost	27	177	0.562	2.810
10		Stimulus Package – Cost/budget	25	186	0.590	2.952
11		Re-work - Cost	7	264	0.838	4.190
12		Variations	8	264	0.838	4.190
13		Wastage - Materials	12	255	0.810	4.048
14		Regular project budget update	38	159	0.505	2.524
15		Cost control method	49	144	0.457	2.286
16		Escalation - material prices	14	251	0.797	3.984
17		Currency prices - Differentiation	34	165	0.524	2.619
18	TIME FEATURES	Time - Site preparation	28	171	0.543	2.714
19		Planned construction time	17	242	0.768	3.841
20		Late deliveries - Percentage	30	171	0.543	2.714
21		Variation orders – Implementation time	9	264	0.838	4.190
22		Defects rectification – Time required	18	241	0.765	3.825
23		Claim approval - Delays	10	260	0.825	4.127
24		Regular payments – Average delays	11	258	0.819	4.095
25		Unavailability of resources	6	264	0.838	4.190
26	QUALITY	Average delay because of closures leading to materials shortage	61	66	0.210	1.048
27		Conformity with specification	21	210	0.667	3.333
28		The absence of a skilled workforce	1	278	0.883	4.413
29		Quality - equipment and raw materials	15	248	0.787	3.937
30		Quality assessment method in company	40	156	0.495	2.476
31	EFFICIENCY	Quality training/meeting	24	195	0.619	3.095
32		Difficulties met during the project	23	198	0.629	3.143
33		New projects rate per annum	41	153	0.486	2.429
34		Relationship between Management & labour	26	185	0.587	2.937
35		Absence rate on Project	13	254	0.806	4.032
36	OWNER	Work sequence with respect to programme	16	246	0.781	3.905
37		Coordination bet. owner and other stakeholders	32	167	0.530	2.651
38		PM – leadership skills	4	274	0.870	4.349
39		Speed and reliability of service	39	158	0.502	2.508
40		Disagreements – Client & project stakeholders	52	125	0.397	1.984
41	COMPLIANCE	Amount of re-works	3	275	0.873	4.365
42		Compliance cost (Authorities)	60	92	0.292	1.460
43		Frequency - Non-compliant cases	58	104	0.330	1.651
44		Regulatory documents – Quality & Readiness	19	214	0.679	3.397
45		Issues – Site conditions	42	153	0.486	2.429
46	EMPLOYEE	Workers' defiance	43	153	0.486	2.429
47		Employment and skill improvement	44	152	0.483	2.413
48		Workers stimulus	59	94	0.298	1.492
49		Sense of belonging to work	5	268	0.851	4.254
50		HOS - Application in organization	57	111	0.352	1.762
51	H&S	Project place - safe to reach	50	144	0.457	2.286
52		Project accident rate – (Reportable ones only)	51	125	0.397	1.984
53		Project Assurance rate	55	119	0.378	1.889
54		Learning (individual practice and former history)	53	121	0.384	1.921
55	LEARNING	Learning (best practice & exp. of others)	46	151	0.479	2.397
56		Work group	56	116	0.368	1.841
57		Appraisal of failures & Provision of solution	20	213	0.676	3.381
58		Quality of Air	54	121	0.384	1.921
59	ENVIRONMENT	Level of noise	47	149	0.473	2.365
60		Trashes	48	147	0.467	2.333
61		Climate condition	45	151	0.479	2.397

The highest 20 performance factors, as identified by the major stakeholders, are as shown in table 4 to 6.

Table 4. 20 highest performance factors as ranked by Clients.

	PERFORMANCE FEATURES	Rank	Weighted total	RII	Item Mean
38	PM – leadership skills	1	248	0.855	4.276
2	Liquidity within company	2	240	0.828	4.138
28	The absence of a skilled workforce	3	239	0.824	4.121
25	Unavailability of resources	4	237	0.817	4.086
11	Re-work - Cost	5	225	0.776	3.879
12	Variations	6	221	0.762	3.810
21	Variation orders – Implementation time	7	220	0.759	3.793
22	Defects rectification – Time required	8	219	0.755	3.776
23	Claim approval - Delays	9	213	0.734	3.672
24	Regular payments – Average delays	10	211	0.728	3.638
13	Wastage - Materials	11	210	0.724	3.621
35	Absence rate on Project	12	207	0.714	3.569
16	Escalation - material prices	13	206	0.710	3.552
29	Quality - equipment and raw materials	14	202	0.697	3.483
36	Work sequence with respect to programme	15	198	0.683	3.414
41	Amount of re-works	16	191	0.682	3.411
19	Planned construction time	17	192	0.662	3.310
49	Sense of belonging to work	18	191	0.659	3.293
44	Regulatory documents – Quality & Readiness	19	190	0.655	3.276
57	Appraisal of failures & Provision of solution	20	190	0.655	3.276

Table 5. 20 highest performance factors as ranked by Consultants.

	PERFORMANCE FEATURES	Rank	Weighted total	RII	Item Mean
38	PM – leadership skills	1	128	0.914	4.571
28	The absence of a skilled workforce	2	126	0.900	4.500
2	Liquidity within company	3	125	0.893	4.464
16	Escalation - material prices	4	124	0.886	4.429
25	Unavailability of resources	5	122	0.871	4.357
35	Absence rate on Project	6	119	0.850	4.250
11	Re-work - Cost	7	118	0.843	4.214
36	Work sequence with respect to programme	8	118	0.843	4.214
57	Appraisal of failures & Provision of solution	9	117	0.836	4.179
49	Sense of belonging to work	10	116	0.829	4.143
41	Amount of re-works	11	116	0.829	4.143
12	Variations	12	115	0.821	4.107
21	Variation orders – Implementation time	13	115	0.821	4.107
22	Defects rectification – Time required	14	114	0.814	4.071
23	Claim approval - Delays	15	113	0.807	4.036
24	Regular payments – Average delays	16	112	0.800	4.000
13	Wastage - Materials	17	112	0.800	4.000
29	Quality - equipment and raw materials	18	111	0.793	3.964
19	Planned construction time	19	108	0.771	3.857
44	Regulatory documents – Quality & Readiness	20	107	0.764	3.821

Table 6. 20 highest performance factors as ranked by Contractors.

	PERFORMANCE FEATURES	Rank	Weighted total	RII	Item Mean
28	The absence of a skilled workforce	1	278	0.883	4.413
2	Liquidity within company	2	276	0.876	4.381
41	Amount of re-works	3	275	0.873	4.365
38	PM – leadership skills	4	274	0.870	4.349
49	Sense of belonging to work	5	268	0.851	4.254
25	Unavailability of resources	6	264	0.838	4.190
11	Re-work - Cost	7	264	0.838	4.190
12	Variations	8	264	0.838	4.190
21	Variation orders – Implementation time	9	264	0.838	4.190
23	Claim approval - Delays	10	260	0.825	4.127
24	Regular payments – Average delays	11	258	0.819	4.095
13	Wastage - Materials	12	255	0.810	4.048
35	Absence rate on Project	13	254	0.806	4.032
16	Escalation - material prices	14	251	0.797	3.984
29	Quality - equipment and raw materials	15	248	0.787	3.937
36	Work sequence with respect to programme	16	246	0.781	3.905
19	Planned construction time	17	242	0.768	3.841
22	Defects rectification – Time required	18	241	0.765	3.825
44	Regulatory documents – Quality & Readiness	19	214	0.679	3.397
57	Appraisal of failures & Provision of solution	20	213	0.676	3.381

From the above tables, it was observed that the performance factors that made the list of 20 were the same but their ranking, which is based on the RII value, was different. Out of the top 5 factors, the following **three** were familiar to all major stakeholders, namely (i) PM – leadership skills, (ii) liquidity within the company and (iii) absence of skilled workforce.

Performance factor 1: COST

The cost factors that stakeholders had in common were;

Liquidity within the company – Situations may arise when payment for works done is being delayed, and in order not to hinder progress, the company may use its funds to make payments to workers and/or suppliers.

Re-works – Occurrence of re-works may be synonymous to shoddy quality, which in turn may result in the company paying, out of its pocket, for materials and labour used for the remedying or re-works. This decreases the profit of the company and also, depending on the extent of reworks, may delay the completion of the project.

Variations – Change orders by Clients/Consultants or unforeseen circumstances, during the construction phase of the project may end up consuming more than the contingency funds allowed on the project. Hence, resulting in the project going over the budget.

Wastage of materials - In Hong Kong, a study by Chu (2004), identified the contribution of the following materials waste to the total project cost: concrete 4%, block work 10%, waste from screeding and plastering 15%, packaging 5% and that of formwork is based on the number of times it is re-used.

Escalation of material prices – This is beyond the control of Contractors. However, if no provision for the escalation of material prices is made in the contract, then the contractor will suffer a loss. In Mauritius, the price escalation is mainly associated with an increase in the amount of petrol as the significant part of the building materials are imported from countries such as South Africa, England, and Egypt among others.

Performance factor 2: TIME

Variation orders (VO) – Implementation Time of variation orders, depending on the degree of complexity of the variations, may impede on the overall duration of the project.

Defects rectifications – Time required to remedy defects, depending on the extent of the flaws, may impact negatively on the overall duration of the project.

Claim approval – On most projects, the approval of claims followed a process and any delays encountered has severe negative impacts on the disbursement of funds, which in turn affects the project duration. For illustration, if the contractor's claim is not approved on time, this means that payment will be delayed and as a consequence supplier will not be paid on time, and future delivery/supply of materials for the project will be jeopardised.

Regular payments – Is linked to the approval of the claims.

Unavailability of resources – Not having the required plant and equipment, as well as the human resources (key personnel) for a project, have negative consequences on the overall duration.

Performance factor 3: QUALITY

The absence of a skilled workforce - Unavailability of qualified/competent staff within the project team is foreseen to impact negatively on the project performance by clients, consultants as well as contractors. The level of education of the workers entering the Mauritian construction to work on the different trades is shallow. It has been observed, through our survey, that most of the workers have studied up to grade six, end of primary school. Currently, the workers are formed on the job site. However, the training they get is not appropriate. It is high time that a proper and rigorous training scheme be explicitly devised for all the different construction trades and be disseminated on the job site itself by qualified trainers. Only after developing the required competencies, will the workers be in a position to execute their tasks to the required standard and quality? An assessment of the developed skillsets is mandatory and each successful worker should either be provided with a certificate of competence or a skill card on which it is stipulated the skill level achieved for his trade. This system, once implemented, will enable future employers to appoint these workers directly for the job for which they are competent and in so doing the quality as well as the overall performance of the project will be sustained.

Quality – equipment and raw materials – All tools, equipment and materials to be used on a project need to satisfy the required minimum quality standards.

Performance factor 4: PRODUCTIVITY

Absence rate on the project – On many projects where the worker is not employed by the contractor, it has been observed that the price of absenteeism is high. For workers engaged in a construction company, fewer absentees are witnessed, which is attributed to the fact that these employees are paid an attendance bonus.

Work sequence concerning the programme - during the construction phase, it is essential that the proposed works methodology (which is expected to be logical and practical) is followed as such an endeavour enables timely completion of the works. This is in line with the results obtained by Samson and Lema (2005), who revealed that the order of tasks to be executed impacts on the efficiency of contractors' delivery outcomes.

Performance factor 5: CLIENT SATISFACTION

The ability of project managers to lead a project has been placed in the 1st place by the participants of the research work. The results of this particular research work concur with the study done by Cheung et al. (2004) where correlations were established between leadership skills of project managers and extent of project success/failure as well as client satisfaction.

Performance factor 6: REGULATORY FACTORS

Quality and availability of regulator documentation have been ranked by all respondents in the 1st position as it upsets the performance of community satisfaction. A similar trend was observed in the study carried out by Samson and Lema (2005).

Performance factor 7: EMPLOYEE FACTORS

“Belonging to work” has been ordered by all stakeholders in number one position because the latter factor typically increases efficiency and performance of building venture, which is coherent with the outcomes of Jha and Iyer (2006). In the Mauritian construction sector, there is a tendency for job contractors as well as sub-contractors to employ the minimum number of workers for a job. Hence, the workers are expected to demonstrate multi-tasking skillsets, which is seldom the case, and as a consequence, the project on the whole suffers.

Performance factor 8: HEALTH & SAFETY FACTORS

This aspect is very significant to all the parties as health and safety factors, within the construction environment, increases project performance globally. This is in line with the observations made by Cheung *et al.* (2004).

Performance factor 9: LEARNING FACTORS

A study by Samson and Lema (2005) confirmed that knowledge acquired from earlier practice distresses the performance of projects. Most of the time clients tend to lean towards the budget rather than other technical issues that are important for the smooth running of the project.

Performance factor 10: ENVIRONMENTAL FACTORS

Climate condition prevailing on site has been ranked in the 1st position by the clients', professionals', and builders' participants as it distresses the efficiency and time performance of the project.

3.1 Degree of Agreement between Responding Groups

The extent to which the different parties to the survey agree or disagree among themselves were determined using Kendall's coefficient of concordance, and the result is presented in table 7.

With Kendall's coefficient of concordance $W = 0.954$ being more significant than 0.5 (Landis and Koch, 1977), the null hypothesis, H_0 is rejected as the degree of agreement among the major stakeholders are statistically significant.

[Null hypothesis (H_0): There is an insignificant degree of agreement among Clients, Contractors and Consultants) and the alternative hypothesis (H_1): There is a statistically significant degree of consensus among owners, contractors and consultants) were tested.]

4. Concluding remarks

A structured questionnaire approach was adopted to capture the perceptions of the major stakeholders operating within the construction industry regarding the performances of smaller projects being executed in small island states.

Table 7. Kendall's coefficient of concordance for stakeholders' survey.

	PERFORMANCE FEATURES	Client	Consultants	Contractors	ΣR	D	D ²
1	Company's market share	61	61	22	144	-51	2601
2	Liquidity within company	2	2	2	6	87	7569
3	Project - Cash flow	32	32	36	100	-7	49
4	Project - Profit rate	29	29	33	91	2	4
5	Project - Overheads	23	23	29	75	18	324
6	Project - design cost	27	27	31	85	8	64
7	Material and equipment cost	31	31	35	97	-4	16
8	Project - labour cost	33	33	37	103	-10	100
9	Project - over time cost	21	21	27	69	24	576
10	Stimulus Package – Cost/budget	24	24	25	73	20	400
11	Re-work - Cost	5	5	7	17	76	5776
12	Variations	6	6	8	20	73	5329
13	Wastage - Materials	11	11	12	34	59	3481
14	Regular project budget update	34	34	38	106	-13	169
15	Cost control method	47	47	49	143	-50	2500
16	Escalation - material prices	13	13	14	40	53	2809
17	Currency prices - Differentiation	30	30	34	94	-1	1
18	Time - Site preparation	22	22	28	72	21	441
19	Planned construction time	17	17	17	51	42	1764
20	Late deliveries - Percentage	25	25	30	80	13	169
21	Variation orders – Implementation time	7	7	9	23	70	4900
22	Defects rectification – Time required	8	8	18	34	59	3481
23	Claim approval - Delays	9	9	10	28	65	4225
24	Regular payments – Average delays	10	10	11	31	62	3844
25	Unavailability of resources	4	4	6	14	79	6241
26	Average delay because of closures - leading to shortage	60	60	61	181	-88	7744
27	Conformity with specification	45	45	21	111	-18	324
28	Absence of skilled workforce	3	3	1	7	86	7396
29	Quality - equipment and raw materials	14	14	15	43	50	2500
30	Quality assessment method in company	36	36	40	112	-19	361
31	Quality training/meeting	50	50	24	124	-31	961
32	Difficulties met during the project	26	26	23	75	18	324
33	New projects rate per annum	37	37	41	115	-22	484
34	Relationship between Management & labour	42	42	26	110	-17	289
35	Absence rate on Project	12	12	13	37	56	3136
36	Work sequence with respect to programme	15	15	16	46	47	2209
37	Coordination bet. owner and other stakeholders	28	28	32	88	5	25
38	PM – leadership skills	1	1	4	6	87	7569
39	Speed and reliability of service	35	35	39	109	-16	256
40	Disagreements – Client & project stakeholders	51	51	52	154	-61	3721
41	Amount of re-works	16	16	3	35	58	3364
42	Compliance cost (Authorities)	59	59	60	178	-85	7225
43	Frequency - Non-compliant cases	57	57	58	172	-79	6241
44	Regulatory documents – Quality & Readiness	19	19	19	57	36	1296
45	Issues – Site conditions	38	38	42	118	-25	625
46	Workers' defiance	39	39	43	121	-28	784
47	Employment and skill improvement	40	40	44	124	-31	961
48	Workers stimulus	58	58	59	175	-82	6724
49	Sense of belonging to work	18	18	5	41	52	2704
50	HOS - Application in organization	56	56	57	169	-76	5776
51	Project place - safe to reach	48	48	50	146	-53	2809
52	Project accident rate – (Reportable ones only)	49	49	51	149	-56	3136
53	Project Assurance rate	54	54	55	163	-70	4900
54	Learning (individual practice and former history)	52	52	53	157	-64	4096
55	Learning (best practice & exp. of others)	43	43	46	132	-39	1521
56	Work group	55	55	56	166	-73	5329
57	Appraisal of failures & Provision of solution	20	20	20	60	33	1089
58	Quality of Air	53	53	54	160	-67	4489
59	Level of noise	44	44	47	135	-42	1764
60	Trashes	46	46	48	140	-47	2209
61	Climate condition	41	41	45	127	-34	1156
					ΣR	ΣD^2	16233
					5673		0
					R =	93	
					0.953		
					W =	82	

According to this study, the major factors found to impede on performances were as listed below;

- 1 PM – leadership skills
- 2 The absence of a skilled workforce
- 3 Liquidity within company
- 4 Escalation - material prices
- 5 Unavailability of resources
- 6 Absence rate on Project
- 7 Re-work - Cost
- 8 Work sequence concerning the programme
- 9 Appraisal of failures & Provision of solution
- 10 The sense of belonging to work
- 11 Amount of re-works
- 12 Variations
- 13 Variation orders – Implementation time
- 14 Defects rectification – Time required
- 15 Claim approval - Delays
- 16 Regular payments – Average delays
- 17 Wastage - Materials
- 18 Quality - equipment and raw materials
- 19 Planned construction time
- 20 Regulatory documents – Quality & Readiness

With Kendall's coefficient of concordance $W = 0.954$ being more significant than 0.5 it was found that the degree of agreement among the major stakeholders was statistically significant and that all the above factors can be used to explain the success or failure of such projects.

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References

- [1] R. Takim and A. Akintoye, "Performance indicators for successful construction project performance," *18th Annual ARCOM Conference*, vol. 2, pp. 545-55, 2002.
- [2] D.K. Chua, Y.C. Kog, and P.K. Loh, "Contract success factors for different project objectives," *Journal of Construction Engineering and Management*, 125(3):142-150, 1999.
- [3] T.R. Puspasari, "*Factors causing poor Performance of Construction Projects.*," An Unpublished Master project Report, submitted to the Faculty of Civil Engineering, University of Technology, Malaysia, 2005.
- [4] D.R. Ogunsemi, "*Predicting the Final Cost of Construction in Nigeria*," *The Quantity Surveyor* 54(4):3-6, 2006.
- [5] H. Yaman, "*A building cost estimation model based on functional elements*," Publication of Istanbul Technical University, A/2(4):73-87, 2007.
- [6] M.Y. Cheng, H.C. Tsai, and E. Sudjono, "*Evolutionary Fuzzy Hybrid Neural Network for Conceptual Cost Estimates in Construction Projects*," 26th International Symposium on Automation and Robotics in construction (ISARC2009): 512-519, 2009.
- [7] M.Y. Cheng, H.C. Tsai, and E. Sudjono, "*Evaluating Subcontractors Performance Using Evolutionary Fuzzy Hybrid Neural network*," *International Journal of Project Management*. 29(2011):249-356, 2011.
- [8] A. Faridi, and S. El-Sayegh, "*Significant factors causing delay in the UAE construction industry*", *Construction Management and Economics*, 24(11): 1167-1176, 2006.
- [9] O.M. Ajayi, O. E. Ogunsami, A. K. Ajayi, and C.M. Ofili, "*Factors Affecting Performance of Contractors on Construction Projects in Lagos State*," *Proceedings of the Construction*,

- Building and Real estate Research Conference of the Royal Institute of Chartered Surveyors, Paris 2-3 September 2010.
- [10] A. Griffith, and J. Headley, “*Management of small buildingworks*,” Construction Management and Economics 16 (6), 703– 709, 1998.
 - [11] T.X. Austin, P.S. Dunston, A.G. Reed, “*Benefits of small projects team initiative*,” Journal of Construction Engineering Management 126 (1), 22–28, 2000.
 - [12] L. Liang, “*Small Project Benchmarking*,” Ph.D Thesis, The University of Texas at Austin, Austin, TX, 2005.
 - [13] L. Briguglio, “*Small Island developing state and their economic Vulnerabilities*”, World Development, Vol.23, No.9, pp 1615-1632, 1995.
 - [14] D.G. Lockhart, D. Drakakis-Smith, J. Schembri, “*The Development Process in Small Island States*”, Routledge, London, 1993.
 - [15] D. Conway, “*Microstates in a macroworld*. In: Klak, T. (Ed.), *Globalization and Neoliberalism in the Caribbean Context*”, Row-man & Littlefield, Oxford, pp. 51–63, 1998.
 - [16] T.N. Slade, “*Special issue on Small Island developing states*”, Natural Resources Forum 23 (3), 185–186, 2002.
 - [17] Commonwealth Secretariat, “*Vulnerability: Small states in Global Society*”, London, 1985.
 - [18] Commonwealth Secretariat, “*Small States: Meeting challenges in the Global Economy*”, Report prepared for the Commonwealth Secretariat World Bank Joint taskforce on small states, Washington, DC, 2000.
 - [19] Statistics Mauritius, “*Population Vital statistics*”, Jan-June 2016, 2016.
 - [20] United Nations, “*Report of the International Meeting to Review the Implementation of the Programme of Actions for the Sustainable Development of Small Island Developing States*”, Port-Louis, Mauritius, 2005.
 - [21] K.N. Jha, “*Construction Project Management: Theory and Practice*”, Pearson India Education Services Pvt. Ltd., 2015.
 - [22] Ugwu, T.C. Haupt, “*Key performance indicators and assessment methods for infrastructure sustainability-a South African construction industry perspective*”, Building and Environment, 42(2), 665-680, 2007.
 - [23] K.N. Jha, C.K. Iyer, “*What attributes should a project coordinator possess?*”, Journal of Construction Management and Economics, 24(9), pp. 977-988, 2006 [Online].
 - [24] P.E.D. Love, and G.D. Holt, “*Construction business performance measurement: the SPM alternative*”. Business Project Management Journal, 6(5): 408-416, 2005.
 - [25] R. Navon, “*Automated project performance control of construction projects*,” Automation of Construction, Vol. 14, Pp. 467-476, 2005.
 - [26] M. Samson, and N.M. Lema, “*Development of construction. Contractor’s performance measurement framework*,” Department of Construction Technology and Management, University of Dares Salaam, Tanzania, 2005.
 - [27] J.A. Kuprenas, “*Project management actions to improve design phase cost performance*”, Journal of Management in Engineering 19(1): 25-32, 2003.
 - [28] S.O. Cheung, H.C. Suen, and K.K. Cheung, “*PPMS: a web-based construction project performance monitoring system*”, Automation in construction, 13(3), 361-376, 2004.
 - [29] Lehtonen Tutu Wegelius, “*Performance measurement in construction logistics*”, International Journal of Production Economics 69: 107-116, 2001.
 - [30] A. Brown, J. Adams, “*Measuring the effect of project management on construction outputs: a new approach*”, International Journal of Project Management 18: 327–335, 2000.
 - [31] Department of the Environment, Transport and the Regions (DETR), “*KPI Report for the Minister for Construction by the KPI Working Group*”, January 2000.
 - [32] M. Dissanayaka, S. Kumaraswamy, and M. Mohan, “*Evaluation of factors affecting time and cost performance in Hong Kong building projects*”, Engineering Construction and Architectural Management. 6. 287 – 298, 1999.
 - [33] K. Karim, and M. Marosszeky, “*Process monitoring for process re-engineering – using key performance indicators*”, International Conference on Construction Process Re-engineering, CPR 99, Sydney, 1999.
 - [34] K. Reichelt, and J. Lyneis, “*The dynamic of project performance: Benchmarking the drivers of cost and schedule overrun*”, European Management Journal 17(2): 135–150, 1999.
 - [35] M. Kendall, Gibbons, *JD Rank correlation methods*. Oxford: Oxford University Press, 1990.